llmXive Automated Review of UI-MOPD: Multi-Platform On-Policy Distillation for Continual GUI Agent Learning

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the llmXive agent-prompt library; each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The paper presents a novel method for cross-platform GUI agents, but several factual claims rely on non-existent or unverified external entities, and there are minor numerical inconsistencies between sections.

First, the data collection pipeline relies on specific teacher models that appear to be hallucinated. The Appendix (Section "Query Generation") and Section 3.1 explicitly state that "Kimi-K2.6" and "Gemini-3.1-Pro" were used to generate queries and trajectories. As of the current public record, neither "Kimi-K2.6" nor "Gemini-3.1-Pro" exist (current versions are typically Kimi 1.5/2.0 and Gemini 1.5/2.0). Citing non-existent models as the source of the training data undermines the reproducibility of the results. The authors must verify the actual model versions used and correct the text and bibliography accordingly.

Second, there is a minor discrepancy in the reported dataset size. The Introduction states the dataset contains "nearly 10K high-quality cross-platform interaction trajectories," while Appendix Table 1 ("Dataset Composition") lists a total of "~11.5K" trajectories. While "nearly 10K" could be interpreted as a conservative lower bound or a rounding error, the specific number 11.5K in the table suggests the Introduction should be updated to "approximately 11.5K" or "over 11K" for consistency.

Finally, the claim of being the "first method" to introduce multi-teacher on-policy distillation (MOPD) into GUI agents is supported by the Related Work section, which notes MOPD is "largely unexplored in GUI agent." However, the authors should ensure that the specific citation for "MOPD" in the Related Work (e.g., `Mimo-v2`, `GLM-5`) accurately reflects that these are general foundation models and not GUI-specific, to avoid any ambiguity about the novelty claim.

These issues are primarily fixable by correcting the model names and aligning the numbers, but the non-existent model names are a significant barrier to verifying the data construction process.

Action items.

- **[writing]** The paper presents a novel method for cross-platform GUI agents, but several factual claims rely on non-existent or unverified external entities, and there are minor numerical inconsistencies between sections. First, the data collection pipeline relies on specific teacher models that appear to be hallucinated. The Appendix (Section "Query Generation") and Section 3.1 explicitly state that "Kimi-K2.6" and "Gemini-3.1-Pro" were used to generate queries and trajectories. As of the current public re

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

The figure provides a clear high-level overview of the data collection pipeline, but contains a spelling error in the 'Bbox Anotation' label and uses low-resolution screenshots that obscure specific interface details.

Figure 2

The figure effectively visualizes a step-by-step GUI agent workflow with clear screenshots and action labels, but the caption is grammatically incomplete and the specific task goal is only visible within a UI element rather than explicitly labeled.

Figure 3

The figure effectively visualizes a mobile GUI agent workflow with clear step-by-step annotations, but the caption is incomplete and the internal reasoning text contains a factual error regarding the file date shown in the screenshot.

Figure 4

The figure provides a clear high-level overview of the training pipeline stages, but the resolution is insufficient to read the mathematical formulas in the bottom panel, and the router's visual representation is slightly abstract.

Figure 5

Figure 5 effectively visualizes the motivation for the proposed method by contrasting naive combination strategies (Model Merge, Mixed SFT) with the UI-MOPD approach. The diagram clearly illustrates the problem of 'Action-Space Conflict' and 'Action Convention Collapse' in the naive methods and how the platform-conditioned routing in UI-MOPD resolves this. The visual elements, labels, and flow are clear and align well with the provided caption.

Action items.

- **[writing]** Figure 1: The text 'Bbox Anotation' contains a spelling error and should be corrected to 'Bbox Annotation'.
- **[writing]** Figure 1: The 'Desktop' and 'Mobile' screenshots are low-resolution and illegible,

making it impossible to verify the specific UI elements or text shown.

- [**writing**] Figure 2: The caption 'Desktop task execution example of .' is grammatically incomplete and missing the specific task name or description.
- [**writing**] Figure 2: The top-left speech bubble contains a specific user prompt ('Can you assist me...'), but the figure lacks a corresponding label or title identifying this as the 'User Task' or 'Goal'.
- [**fatal**] Figure 3: The caption 'Mobile task execution example of .' is incomplete and grammatically broken, failing to describe the specific task shown (replying to a gourmet user about Moussaka).
- [**science**] Figure 3: The 'Structured Chain-of-Thought' box contains a factual hallucination; it states the file 'waiver.jpg' is listed 'Jun 2 2026', but the screenshot clearly shows the date as 'Jun 2, 2025'.
- [**writing**] Figure 3: The top-level instruction bubble is cluttered with a cartoon avatar and excessive whitespace, which distracts from the scientific content of the figure.
- [**writing**] Figure 4: The 'Reverse KL Computation' panel contains mathematical notation (e.g., log (y_t|h_t)) that is illegible due to low resolution, making the specific formula unreadable.
- [**writing**] Figure 4: The 'Router' component is depicted with a speaker icon and dotted lines, but the visual metaphor for 'platform-conditioned routing' is ambiguous without explicit labels on the input/output paths.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The paper is generally well-written and avoids excessive in-group slang, but it relies on a few undefined symbols and subfield-specific acronyms that would stall a competent reader from an adjacent field (e.g., a researcher in NLP or Computer Vision who is not deeply embedded in the specific RLHF/GUI-agent sub-ecosystem).

The primary issue is the introduction of mathematical notation without immediate definition. In Section 2.1, the symbol appears in Equation 3 as an "adaptive KL mask." The text describes its function (removing penalty when reward is sufficient) but never explicitly defines its domain (is it a scalar per sample? a vector per token? a function of the group?). A reader cannot parse the equation's mechanics without guessing. Similarly, the "K3 estimator" is named and used in Equation 5 without a definition or citation. While the formula is provided, the name "K3" implies a specific known method or a novel contribution that requires a one-sentence operational definition (e.g., “a low-variance estimator based on...”) to be accessible.

Additionally, the acronym “DAPO” is used in the hyperparameter table (Section 3.1) without expansion. While “GRPO” is becoming more standard, “DAPO” is not universally recognized outside specific RLHF circles. Expanding this at first use is a trivial fix that significantly improves accessibility. Finally, the notation `f_a` in the reward design section relies on an implicit list of “dimensions” that is only partially enumerated in the text; a formal definition or reference to a complete schema would prevent ambiguity for readers trying to reproduce the reward calculation.

These are minor, text-only fixes that do not alter the scientific content but are necessary to ensure the paper is self-contained for the target audience.

Action items.

- [**writing**] Section 2.1 (Method): The symbol is introduced in Eq. 3 as an ‘adaptive KL mask’ but is never defined in the surrounding text. A reader cannot determine if it is a scalar, a vector, or a function of the rollout. Define explicitly where it first appears (e.g., ‘where is a binary mask...’).
- [**writing**] Section 2.1 (Method): The term ‘K3 estimator’ is used in the paragraph header and Eq. 5 without a definition or citation. While ‘KL’ is standard, ‘K3’ appears to be specific to this work or a very niche subfield. Add a brief clause explaining what the K3 estimator is (e.g., ‘a variance-reduced token-level KL estimator’) or cite the source.
- [**writing**] Section 2.3 (Reward Design): The variable `f_a` is introduced as ‘the fraction of matched dimensions’ but the specific dimensions for each action type (e.g., coordinate inclusion, key set equality) are listed as examples rather than a formal definition. An adjacent-field reader cannot verify the calculation without guessing the exact set of dimensions. Explicitly list the dimension set or reference a table/appendix where the full schema is defined.
- [**writing**] Appendix A (Dataset Construction): The term ‘AndroidControl*’ is used repeatedly with a star superscript. While the text mentions it is an ‘evaluated subset’, the specific criteria for inclusion/exclusion (beyond ‘781 trajectories’) are not defined in the main text or this section. Define the selection criteria for the star subset at first use to ensure reproducibility.
- [**writing**] Section 3.1 (Experimental Setup): The acronym ‘DAPO’ appears in the hyperparameter table (‘GRPO-based DAPO objective’) without being expanded in the text. While ‘GRPO’ is a known method, ‘DAPO’ is not standard across the broader ML field. Expand ‘DAPO’ at first use (e.g., ‘Direct Advantage Policy Optimization’) or clarify its relationship to GRPO.

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_logical_consistency.md`

The paper's argument structure is generally sound, with the proposed method (UI-MOPD) logically addressing the stated problem of platform-specific behavioral mixing. The premises regarding the scarcity of cross-platform data and the risks of naive merging are well-established, and the conclusion that platform-conditioned distillation mitigates these issues follows logically from the described mechanism.

However, there are three specific logical and consistency gaps that require attention:

1. **Redundant Narrative in Appendix:** In `Sections/Appendix.tex`, the "Query Generation" paragraph appears twice in immediate succession. The first version is generic, while the second adds specific environment details. This duplication breaks the logical flow of the data construction narrative, presenting two slightly different versions of the same step as if they were distinct phases. This should be merged into a single, precise description.

2. **Inconsistent Dataset Scale:** The Introduction (Section 1) claims the constructed dataset contains "nearly 10K" trajectories. In contrast, the Appendix (Section A) and Table 1 explicitly state the total is "~11.5K" trajectories. While both are rounded figures, a 15% discrepancy in the primary dataset size claim creates ambiguity about the scale of the contribution. The Introduction should be updated to reflect the more precise figure provided in the data composition section to ensure consistency.

3. **Label Collision:** The Appendix defines a figure with `\label{fig:intro}` and caption "Overview of Unified Cross-Platform Data Collection Harness." However, the main body (Section 1) already uses `\label{fig:intro}` for the "Motivation of UI-MOPD" figure. This is a technical logical error in the document structure that will cause the LaTeX compiler to reference the wrong figure when `fig:intro` is cited, breaking the internal consistency of the document's cross-references. The appendix label must be renamed.

These issues are primarily presentational and structural but affect the internal consistency and readability of the argument.

Action items.

- [**writing**] In Appendix A, the 'Query Generation' paragraph is duplicated verbatim with slightly different phrasing. The first instance mentions Kimi-K2.6 and Gemini-3.1-Pro generically, while the second adds specific environment names (OSWorld, MobileWorld). This redundancy creates a logical break in the narrative flow. Merge these into a single, coherent paragraph.

- [**writing**] Section 1 (Introduction) states the dataset contains 'nearly 10K' trajectories, while Appendix A (Dataset Construction) and Table 1 state '~11.5K' trajectories. While both are approximations, the discrepancy (10K vs 11.5K) is significant enough to cause confusion regarding the dataset scale. Align the numbers in the Introduction to match the specific value in the Appendix/Tables (e.g., 'approximately 11.5K').

- [**writing**] In Appendix A, Figure 1 is labeled `fig:intro` and captioned 'Overview of Unified Cross-Platform Data Collection Harness.' However, `fig:intro` is already used in the main body (Section 1) for the 'Motivation of UI-MOPD' figure. This label collision will cause LaTeX to generate incorrect cross-references (e.g., Section 1's figure will point to the harness diagram). Rename the appendix figure label to `fig:harness_overview`.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_overreach.md`

The paper makes several broad claims regarding novelty and generalization that exceed the specific scope of the reported experiments.

First, the claim of being the "first method" to use multi-teacher on-policy distillation (MOPD) for GUI agents (Abstract, Introduction) is technically imprecise. While the application to GUI agents may be novel, the Related Work section explicitly cites DeepSeek-V4 and Nemotron-Cascade 2 as using MOPD for foundation model post-training. The phrasing should be tightened to "the first to apply MOPD to the specific problem of cross-platform GUI agent continual learning" to avoid implying the underlying distillation technique is a new invention.

Second, the rhetoric regarding "catastrophic forgetting" and "continual adaptation" (Abstract, Conclusion) is stronger than the evidence supports. The paper demonstrates improved performance on two static benchmarks (OSWorld and MobileWorld) compared to baselines. However, it does not present a true continual learning experiment where the model is trained sequentially on Platform A, then Platform B, and tested on A to measure forgetting. The claim that the method "mitigates catastrophic forgetting" is an inference drawn from the static comparison, not a direct measurement of retention over time. The text should be hedged to reflect that the method *preserves* performance on the tested benchmarks rather than definitively solving the forgetting problem in a continual setting.

Finally, the conclusion suggests the method provides a path toward agents that can "continually adapt to diverse digital environments." The experimental validation is strictly confined to desktop and mobile platforms. There is no data for web, tablet, or other interface types. The scope of "diverse digital environments" should be explicitly limited to "desktop and mobile environments" in the abstract and conclusion to match the experimental reality.

Action items.

- [**writing**] Abstract/Intro claim UI-MOPD is 'the first method' to use MOPD for GUI agents. Related Work cites DeepSeek-V4/Nemotron using MOPD. Qualify to 'first to apply MOPD to cross-platform GUI continual learning' to avoid implying the technique itself is novel.
- [**writing**] Abstract/Conclusion claim the method 'mitigates catastrophic forgetting' and enables 'continual adaptation.' Experiments only show static benchmark scores (OSWorld/MobileWorld) without a sequential training/forgetting metric. Hedge to 'preserves performance on tested benchmarks' rather than claiming to solve forgetting.

- [**writing**] Conclusion claims applicability to 'diverse digital environments.' Experiments are limited to desktop and mobile. Narrow the claim to 'desktop and mobile environments' in the abstract and conclusion to match the tested scope.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_safety_ethics.md`

This paper presents a method for training cross-platform GUI agents using multi-teacher on-policy distillation. The research involves collecting interaction trajectories from desktop and mobile environments, training teacher models, and distilling knowledge into a student model.

From a safety and ethics perspective, the work is low-risk. The data collection harness described in the Appendix (Section `app:data_collection_harness`) utilizes synthetic query generation and automated trajectory collection within controlled benchmark environments (OSWorld, MobileWorld, AndroidWorld). The paper explicitly states that trajectories are filtered for quality and success, and no human-subjects data, personal information, or sensitive user interactions are involved in the training or evaluation process. The datasets constructed (Uni-GUI, AndroidControl*) are derived from these synthetic or public benchmark sources, and the paper does not release any raw data containing PII.

The method itself (GUI automation) is a standard area of research in the field. While GUI agents have potential dual-use applications (e.g., automating malicious tasks), the paper focuses on improving task success rates and cross-platform generalization for benign automation. It does not describe capabilities that lower the barrier to specific high-harm activities (such as generating exploits, biological synthesis, or targeted disinformation) beyond the general capabilities of the underlying foundation models (Qwen3-VL). The paper does not claim to bypass security controls or operate covertly.

There are no missing disclosures regarding human subjects, consent, or data licensing that would constitute a safety violation. The use of public benchmarks and synthetic data generation avoids the ethical pitfalls associated with scraping private user data. Consequently, no specific safety or ethics action items are required.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The paper presents a compelling method for cross-platform GUI agents, but the experimental design currently leaves open several alternative explanations for the reported gains.

First, the stability of the results is unverified. Table 2 reports a 5.6 percentage point improvement on MobileWorld (12.0% vs 6.4% for Mixed-SFT) and a 3.2 point gain on OSWorld. However, the paper reports these as single-point estimates without any indication of variance, standard deviation, or the number of random seeds used. In reinforcement learning and agent benchmarks, performance can fluctuate significantly between seeds due to the stochastic nature of rollout generation and environment interaction. A gap of this magnitude could plausibly arise from a single lucky seed rather than a robust algorithmic improvement. To establish the claim that UI-MOPD is genuinely superior, the authors must report results averaged over at least 3-5 independent training runs with seeds, providing mean ± standard deviation.

Second, the attribution of the gain to the specific "multi-teacher" mechanism is not fully isolated. The paper claims the improvement comes from "platform-conditioned distillation" preventing behavioral mixing. However, the primary comparison in Table 2 is against "Mixed-SFT" and "Model Merge." It is possible that the improvement stems simply from the use of on-policy reinforcement learning (the DAPO/GRPO objective) rather than the specific multi-teacher routing. A critical control experiment is missing: a "Single-Teacher On-Policy Distillation" baseline where the student is trained with the same RL framework but distills from only one teacher (or a merged teacher) without platform-conditioned routing. Without this ablation, the reader cannot distinguish whether the benefit comes from the *multi-teacher* aspect or the *on-policy* aspect of the method.

Finally, there is a potential confusion in the interpretation of the teacher-student comparison in Section 4.3. The text states UI-MOPD "outperforms the 32B base model on MobileWorld," citing 12.0% vs 9.4%. However, Table 3 explicitly lists the "Mobile Teacher, 32B" achieving 16.2% on MobileWorld, which is higher than the student's 12.0%. The student underperforms the specialized teacher, which is expected, but the text's phrasing suggests a comparison against the 32B *base* (untrained) model or creates ambiguity about whether the student has surpassed the expert. Clarifying this comparison is essential to ensure the claim of "effective transfer" is not overstated.

Action items.

- [**writing**] The paper presents a compelling method for cross-platform GUI agents, but the experimental design currently leaves open several alternative explanations for the reported gains. First, the stability of the results is unverified. Table 2 reports a 5.6 percentage point improvement on MobileWorld (12.0% vs 6.4% for Mixed-SFT) and a 3.2 point gain on OSWorld. However, the paper reports these as single-point estimates without any indication of variance, standard deviation, or the number of random seed

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

The statistical treatment of the experimental results in this paper is currently insufficient to support the inferential claims made. The primary issue is the reporting of point estimates (e.g., "38.2% success rate") without any accompanying measure of uncertainty (standard deviation, standard error, or confidence intervals) or the number of independent training seeds used. In reinforcement learning and agent training, performance is inherently stochastic; a single run or an unreported average can easily be an outlier. Without variance across seeds, it is impossible to determine if the reported improvements over baselines (e.g., the 4.3 point gain on OSWorld) are statistically robust or merely noise.

Furthermore, the paper frequently uses the term "significantly" (e.g., in the Abstract and Section 4.3) to describe improvements, yet no formal hypothesis tests (such as paired t-tests or bootstrap tests) are performed, and no p-values are reported. The claim of "statistical significance" requires a test statistic and a threshold, neither of which are present. The authors must either compute these metrics from seed-level data or rephrase their claims to reflect observed differences without statistical inference.

Finally, the comparison involves multiple baselines across multiple benchmarks. The paper highlights the best-performing cells in Table 1 without applying any correction for multiple comparisons (e.g., Bonferroni or Benjamini-Hochberg). This increases the family-wise error rate, meaning some of the highlighted "best" results may be false positives. A correction method should be applied, or the multiplicity of tests should be explicitly acknowledged. These are not minor formatting issues but fundamental gaps in the statistical evidence required to validate the paper's core claims.

Action items.

- [**science**] Table 1 and Section 4.3 report single-point success rates (e.g., 38.2%, 12.0%) without any measure of uncertainty (SD, SE, or CI) or number of seeds. In RL-based agent training, performance is highly stochastic. Report mean ± SD over at least 3 independent training seeds for all main results to distinguish stable improvements from random variance.
- [**writing**] The abstract and Section 4.3 claim UI-MOPD is 'significantly better' or shows 'substantial improvements' over baselines (e.g., 12.7% and 55.8% relative gains) but provide no statistical hypothesis tests (e.g., paired t-test, bootstrap) or p-values. Without variance estimates or formal tests, these 'significant' claims are unsupported. Either run statistical tests on seed-level results or rephrase to 'observed improvement' without statistical inference.
- [**writing**] Table 1 compares UI-MOPD against 4 baselines across 2 benchmarks (8 pairwise comparisons) and highlights the best results. No correction for multiple comparisons (e.g., Bonferroni, Holm, or FDR) is mentioned. With multiple tests, the risk of false positives increases. Apply a correction method or explicitly acknowledge the uncorrected multiplicity when interpreting 'best' results.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The paper is generally well-structured and readable, with clear logical flow between sections. The abstract effectively summarizes the problem, method, and results. However, there are specific areas where the prose creates friction for the reader.

The most significant issue is in the Appendix, Section 2 ("Unified Cross-Platform Data Collection Harness"). There are two consecutive paragraphs both titled "Query Generation." The first paragraph describes generating queries using Kimi-K2.6 for desktop and Gemini-3.1-Pro for mobile. The second paragraph repeats this information almost verbatim, adding specific environment names (OSWorld, MobileWorld, AndroidWorld) but failing to clarify that it is a revision or expansion of the previous point. This duplication forces the reader to re-read and wonder if these are distinct steps or an editing error. These should be merged into a single, cohesive paragraph that introduces the environments and models together.

Additionally, within the same section, the "Trajectory Cleaning" paragraph uses a sequential list ("First," "Second," "Third," "Fourth") but then immediately follows with a paragraph starting "Finally, we use Gemini-3.1-Pro..." This breaks the logical flow of the enumeration. The "Finally" paragraph describes a specific filtering step (sub-task adjudication) that logically belongs within the cleaning sequence, not as a separate post-script. Integrating this step into the numbered list or rephrasing the transition would improve clarity.

Finally, there is a minor inconsistency in the dataset size reporting. The Introduction claims the dataset contains "nearly 10K" trajectories, while the Appendix specifies "11.5K trajectories." While close, scientific writing benefits from precise consistency. Aligning these figures (e.g., "approximately 11.5K" in the Introduction) would eliminate even this small source of reader hesitation.

Overall, the paper's writing quality is high, but these specific structural and redundancy issues prevent it from being perfectly frictionless.

Action items.

- [**writing**] The paper is generally well-structured and readable, with clear logical flow between sections. The abstract effectively summarizes the problem, method, and results. However, there are specific areas where the prose creates friction for the reader. The most significant issue is in the Appendix, Section 2 ("Unified Cross-Platform Data Collection Harness"). There are two consecutive paragraphs both titled "Query Generation." The first paragraph describes generating queries using Kimi-K2.6 for deskt